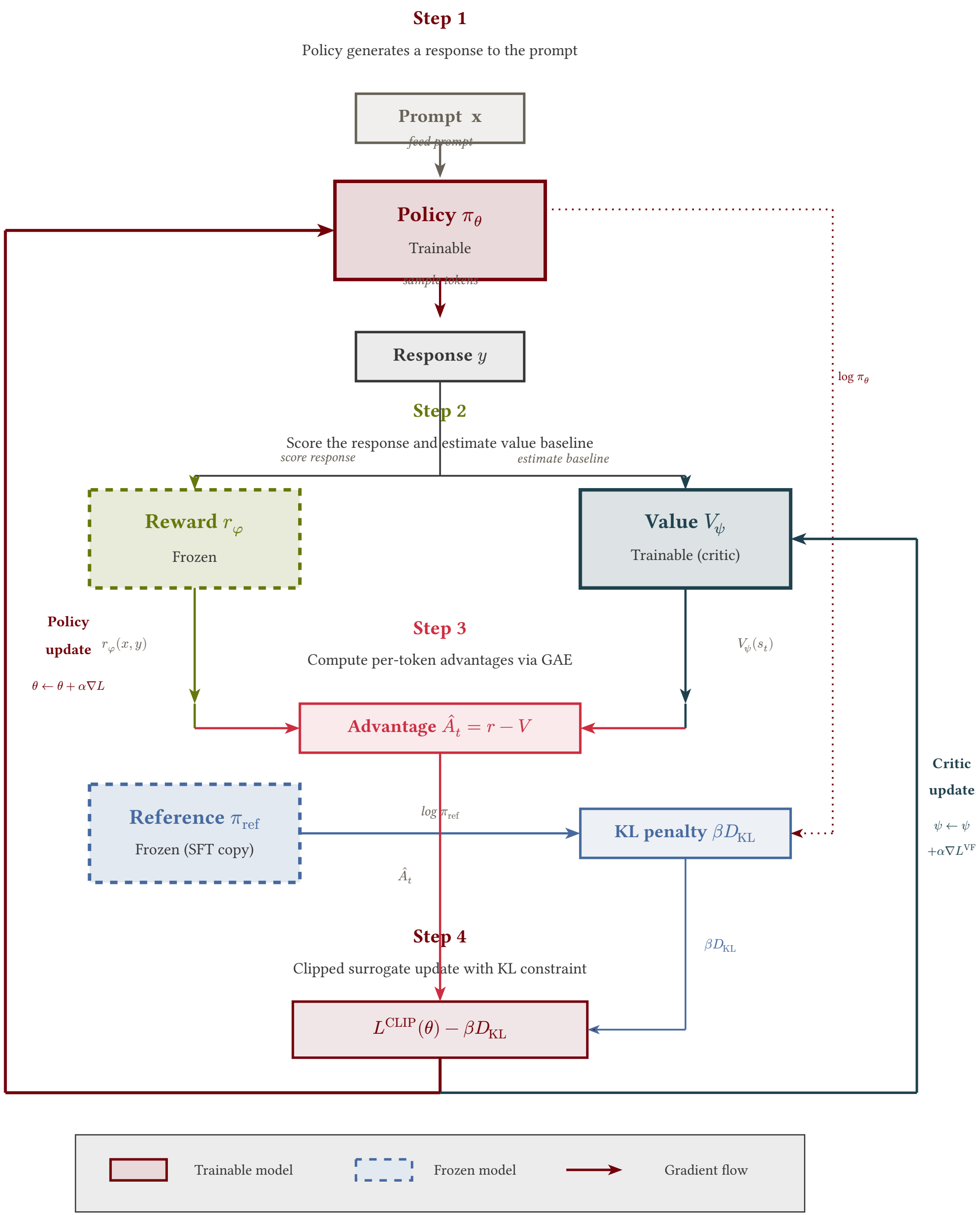

Step 1
Policy generates a response to the prompt
Prompt x
feed prompt
Policy π_θ
Trainable
sample tokens
Response y
$\log \pi_\theta$
Step 2
Score the response and estimate value baseline
score response
estimate baseline
Reward r_φ
Frozen
Value V_ψ
Trainable (critic)
Policy update
$r_\varphi(x, y)$
$\theta \leftarrow \theta + \alpha \nabla L$
Step 3
Compute per-token advantages via GAE
$V_\psi(s_t)$
Advantage $\hat{A}_t = r - V$
Critic update
$\psi \leftarrow \psi$
$+\alpha \nabla L^{\mathrm{VF}}$
Reference π_{ref}
Frozen (SFT copy)
$\log \pi_{\mathrm{ref}}$
KL penalty βD_{KL}
$\hat{A}_t$
βD_{KL}
Step 4
Clipped surrogate update with KL constraint
$L^{\mathrm{CLIP}}(\theta) - \beta D_{\mathrm{KL}}$
Trainable model
Frozen model
Gradient flow